

Gosford High School

Year 11

2008

Preliminary Higher School Certificate

Mathematics

Assessment Task 3

Time Allowed - 65 minutes

Remember to start each new question on a new page

Students must answer questions using a blue/black pen and/or a sharpened B or HB pencil.

Approved scientific calculators may be used

Students need to be aware that

- * 'bald' answers may not gain full marks.

- * untidy and/or poorly organised solutions may not gain full marks.

QUESTION 1 (12 marks)

- (i) Plot the points $A(7,3)$, $B(-2,2)$ and $C(3,-2)$ on a number plane – use approximately $\frac{1}{4}$ of an A4 page and label your diagram clearly. (2)
- (ii) Find the coordinates of the midpoint of AC . (1)
- (iii) Find the gradient of the line AB . (1)
- (iv) Show that the line l drawn parallel to AB and passing through C has equation $x - 9y - 21 = 0$. (2)
- (v) Find the coordinates of the point D where the line l intersects the x axis. (1)
- (vi) Find the perpendicular distance from the line l to the point A . (2)
- (vii) Find the area of the triangle ABD . (3)

QUESTION 2 (13 marks)

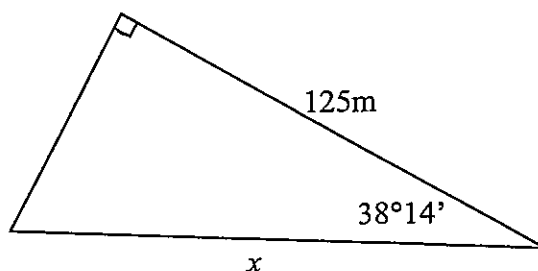
- (a) What is the gradient of a line drawn perpendicular to the line $6x + 2y - 7 = 0$ (2)
- (b) Find the equation of the line k which passes through the point $(-\sqrt{3}, 2)$ and makes an angle of 150° with the positive direction of the x axis. (2)
- (c) Find the point of intersection of the lines $2x - 3y = 5$ and $x + 5y = 4$ (2)
- (d) The lines with equations $5x + 2y = 19$ and $4x + ay = b$ are perpendicular and intersect at the point $(3, 2)$. Find the values of ' a ' and ' b '. (3)
- (e) The equation of any line passing through the point of intersection of the lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ has the form $(a_1x + b_1y + c_1) + k(a_2x + b_2y + c_2) = 0$, where k is a constant.
- (i) Find the value of the constant k required to find the equation of the line passing through the point $(1, -2)$ and the point of intersection of the lines $3x + 4y - 2 = 0$ and $5x - 2y - 12 = 0$. (2)
- (ii) Hence find the equation of the line passing through the point $(1, -2)$ and the point of intersection of the lines $3x + 4y - 2 = 0$ and $5x - 2y - 12 = 0$ (2)

QUESTION 3

(13 marks)

(a)

Find the value of x correct to 3 significant figures



(3)

(b) Solve $\tan x = 2.36$ for $180^\circ \leq x \leq 360^\circ$, writing your answer(s) to the nearest degree.

(1)

(c) Solve $\sin A = -\frac{\sqrt{3}}{2}$ for $0^\circ \leq A \leq 360^\circ$

(2)

(d) If $\tan K = -\frac{8}{15}$ and $0^\circ \leq K \leq 270^\circ$ find the exact value of $\sin K$.

(2)

(e) Write the exact value of

(i) $\cot(300^\circ)$

(1)

(ii) $\cos(-45^\circ)$

(1)

(f) A ship leaves port A and sails 30km due North to port B.

The ship then sails 50km due West to port C.

(i) Draw a neat diagram representing the given information.

(1)

(ii) Find $\angle BAC$ to the nearest degree.

(1)

(iii) Hence find the bearing of port A from port C.

(1)

QUESTION 4**(12 marks)**

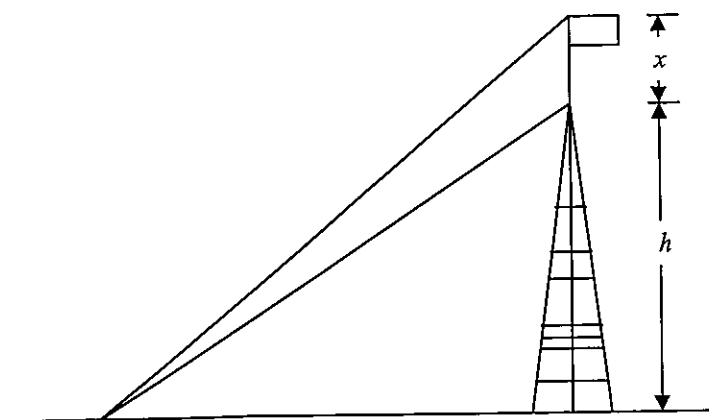
(a) Draw the graph of $y = \sin x$ for $0^\circ \leq x \leq 360^\circ$, labelling all essential features. (2)

(b) Simplify $\frac{\sin(180 + \alpha)}{\cos(90 - \alpha)}$ (2)

(c) Solve $|\cos \theta| = 1$ for $0^\circ \leq \theta \leq 360^\circ$ (2)

(d) Simplify $\frac{\tan \theta + \cot \theta}{\operatorname{cosec} \theta \cdot \sec \theta}$ (2)

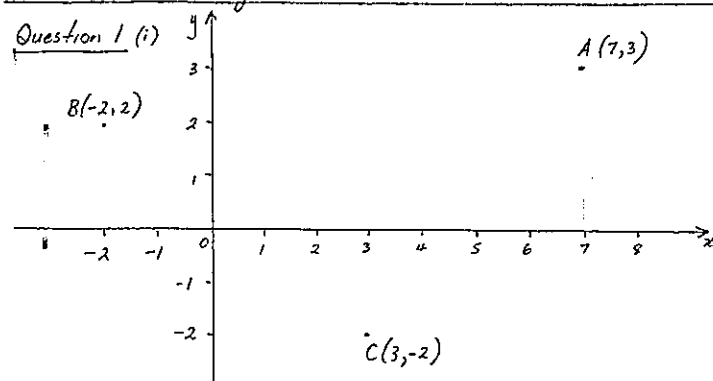
(e)



A flagpole x metres high stands on top of a vertical tower. From a point on the ground, the angles of elevation of the top and bottom of the flagpole are β° and α° respectively. Show that the height of the tower, h m, is given by

$$h = \frac{x \tan \alpha^\circ}{\tan \beta^\circ - \tan \alpha^\circ}. \quad (4)$$

Question 1 (i)



ii) Midpoint of AC = $\left(\frac{7+3}{2}, \frac{3-2}{2}\right)$ using $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$
 $= \left(5, \frac{1}{2}\right)$

iii) Gradient of AB = $\frac{3-2}{7-(-2)}$ using $m = \frac{y_2-y_1}{x_2-x_1}$
 $m_{AB} = \frac{1}{9}$

iv) Equation of L is :-
 $y+2 = \frac{1}{9}(x-3)$ using $y-y_1 = m(x-x_1)$
 $9y+18 = x-3$
 $x-9y-21=0$

Question 2

a) $6x+2y-7=0$
 $2y = -6x+7$
 $y = -3x + \frac{7}{2}$

$\therefore$ Gradient of perpendicular line is $\frac{1}{3}$

b) Gradient of line k = $\tan 150^\circ$
 $= -\frac{1}{\sqrt{3}}$

Equation of line k is
 $y-2 = -\frac{1}{\sqrt{3}}(x+\sqrt{3})$
 $y\sqrt{3}-2\sqrt{3} = -x-\sqrt{3}$
 $x+y\sqrt{3} = \sqrt{3}$

c) $x = 4-5y \rightarrow 2x-3y = 5$
 $2(4-5y)-3y = 5$
 $8-10y-3y = 5$
 $3 = 13y$
 $y = \frac{3}{13}$
 $x = 4 - \frac{15}{13}$
 $x = 2\frac{11}{13}$

Point of Intersection is $\left(2\frac{11}{13}, \frac{3}{13}\right)$

v) L intersects x axis when $y=0$

$\therefore x-21=0$
 $x=21$

D has coordinates (21,0)

vi) $d = \left| \frac{1(7)+(-9)(3)-21}{\sqrt{1^2+(-9)^2}} \right|$ using $d = \left| \frac{ax_1+by_1+c}{\sqrt{a^2+b^2}} \right|$

$d = \left| \frac{7-27-21}{\sqrt{82}} \right|$ and $a=1$ $x_1=7$
 $b=-9$ $y_1=3$
 $c=-21$

$d = \left| \frac{-41}{\sqrt{82}} \right|$

$d = \frac{41}{\sqrt{82}}$

vii) $d_{AB} = \sqrt{(7+2)^2 + (3-2)^2}$
 $= \sqrt{82}$

Area of $\triangle ABD = \frac{1}{2}bh$
 $= \frac{1}{2} \times \sqrt{82} \times \frac{41}{\sqrt{82}}$
 $= 20.5 \text{ sq units}$

d) (3,2) satisfies $4x+ay=b$
 $12+2a=b \dots (1)$

Gradient of $5x+2y=19$ is $-\frac{5}{2} = m_1$

Gradient of $4x+ay=b$ is $-\frac{4}{a} = m_2$

Since $\perp$, $m_1 \times m_2 = -1$

$\frac{20}{2a} = -1$

$20 = -2a$

$a = -10$

$\therefore b = 12-20$ using (1)

$b = -8$

$\therefore a = -10, b = -8$

e) i) Equation of required line is

$(3x+4y-2)+k(5x-2y-12)=0$

(1,-2) satisfies

$\therefore (3-8-2)+k(5+4-12)=0$

$-7-3k=0$

$k = -\frac{7}{3}$

e) ii) Equation required is

$$(3x + 4y - 2) - \frac{7}{3}(5x - 2y - 12) = 0$$

$$9x + 12y - 6 - 35x + 14y + 84 = 0$$

$$-26x + 26y + 78 = 0$$

$$x - y - 3 = 0$$

Question 3

a) $\cos 38^\circ 14' = \frac{125}{x}$

$$x = \frac{125}{\cos 38^\circ 14'}$$

$$x = 159 \text{ m}$$

b) $x = 180^\circ + \tan^{-1}(2.36)$

$$= 180^\circ + 67.03^\circ$$

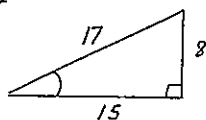
$$= 247^\circ \text{ (to the nearest degree)}$$

c) $A = 180^\circ + 60^\circ, 360^\circ - 60^\circ$

$$= 240^\circ, 300^\circ$$

d) K lies in 2nd Quadrant

$$\therefore \sin K = \frac{8}{17}$$



b) $\frac{\sin(180^\circ + \alpha)}{\cos(90^\circ - \alpha)} = \frac{-\sin \alpha}{\sin \alpha}$

$$= -1$$

c) $|\cos \theta| = 1$

$$\therefore \cos \theta = \pm 1$$

$$\therefore \theta = 0^\circ, 180^\circ, 360^\circ$$

d) $\frac{\tan \theta + \cot \theta}{\operatorname{cosec} \theta \cdot \sec \theta} = \frac{\frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta}}{\frac{1}{\sin \theta} \cdot \frac{1}{\cos \theta}}$

(multiply top & bottom by $\sin \theta \cos \theta$)

$$= \frac{\sin^2 \theta + \cos^2 \theta}{1}$$

$$= \frac{1}{1}$$

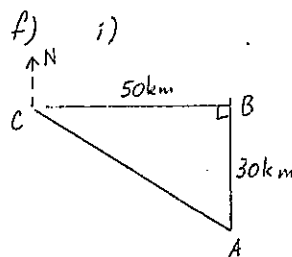
$$= 1$$

e) i) $\cot 300^\circ = \frac{1}{\tan 300^\circ}$

$$= \frac{1}{-\tan 60^\circ}$$

$$= -\frac{1}{\sqrt{3}}$$

ii) $\cos(-45^\circ) = \cos 45^\circ = \frac{1}{\sqrt{2}}$



ii) $\tan \hat{BAC} = \frac{50}{30}$

$$\hat{BAC} = \tan^{-1}\left(\frac{5}{3}\right)$$

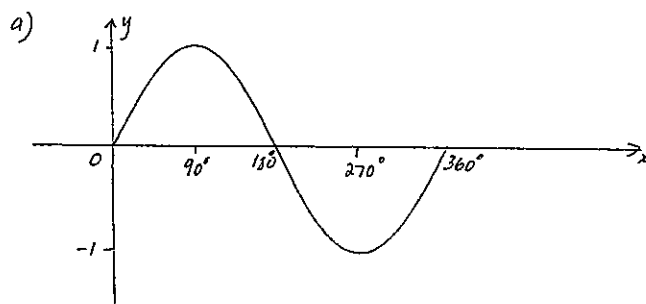
$$= 59^\circ$$

$$\angle NAC = 180^\circ - 59^\circ$$

$$= 121^\circ$$

$\therefore$ Bearing of A from C is 121° N

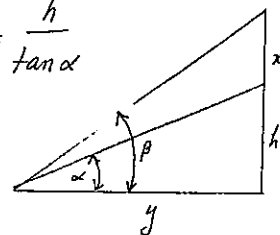
Question 4



e) $\tan \alpha = \frac{h}{y} \rightarrow y = \frac{h}{\tan \alpha}$

$$\tan \beta = \frac{h+x}{y}$$

$$\therefore \tan \beta = \frac{h+x}{\frac{h}{\tan \alpha}}$$



$$\tan \beta = \frac{h \tan \alpha + x \tan \alpha}{h}$$

$$h \tan \beta = h \tan \alpha + x \tan \alpha$$

$$h \tan \beta - h \tan \alpha = x \tan \alpha$$

$$h(\tan \beta - \tan \alpha) = x \tan \alpha$$

$$h = \frac{x \tan \alpha}{\tan \beta - \tan \alpha}$$

e) ii) Equation required is

$$(3x + 4y - 2) - \frac{7}{3}(5x - 2y - 12) = 0$$

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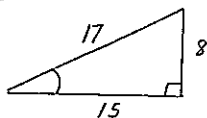
$$\begin{aligned} a) \cos 38^\circ 14' &= \frac{125}{x} \\ x &= \frac{125}{\cos 38^\circ 14'} \\ x &= 159 \text{ m} \end{aligned}$$

$$\begin{aligned} b) z &= 180^\circ + \tan^{-1}(2.36) \\ &= 180^\circ + 67.03^\circ \\ &= 247^\circ \text{ (to the nearest degree)} \end{aligned}$$

$$\begin{aligned} c) A &= 180^\circ + 60^\circ, 360^\circ - 60^\circ \\ &= 240^\circ, 300^\circ \end{aligned}$$

d) K lies in 2nd Quadrant

$$\therefore \sin K = \frac{8}{17}$$



$$\begin{aligned} b) \frac{\sin(180^\circ + \alpha)}{\cos(90^\circ - \alpha)} &= \frac{-\sin \alpha}{\sin \alpha} \\ &= -1 \end{aligned}$$

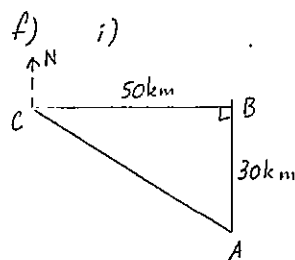
$$\begin{aligned} c) |\cos \theta| &= 1 \\ \therefore \cos \theta &= \pm 1 \end{aligned}$$

$$\therefore \theta = 0^\circ, 180^\circ, 360^\circ$$

$$\begin{aligned} d) \frac{\tan \theta + \cot \theta}{\operatorname{cosec} \theta \cdot \sec \theta} &= \frac{\frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta}}{\frac{1}{\sin \theta} \cdot \frac{1}{\cos \theta}} \\ &\quad \left(\text{multiply top \& bottom by } \sin \theta \cos \theta \right) \\ &= \frac{\sin^2 \theta + \cos^2 \theta}{1} \\ &= \frac{1}{1} \\ &= 1 \end{aligned}$$

$$\begin{aligned} e) i) \cot 300^\circ &= \frac{1}{\tan 300^\circ} \\ &= \frac{1}{-\tan 60^\circ} \\ &= -\frac{1}{\sqrt{3}} \end{aligned}$$

$$\begin{aligned} ii) \cos(-45^\circ) &= \cos 45^\circ \\ &= \frac{1}{\sqrt{2}} \end{aligned}$$

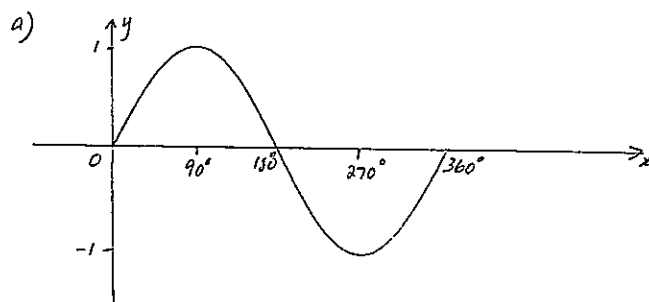


$$\begin{aligned} ii) \tan \hat{BAC} &= \frac{50}{30} \\ \hat{BAC} &= \tan^{-1}\left(\frac{5}{3}\right) \\ &= 59^\circ \end{aligned}$$

$$\begin{aligned} \angle NAC &= 180^\circ - 59^\circ \\ &= 121^\circ \end{aligned}$$

$\therefore$ Bearing of A from C is 121° N

Question 4



$$e) \tan \alpha = \frac{h}{y} \rightarrow y = \frac{h}{\tan \alpha}$$

$$\tan \beta = \frac{h+x}{y}$$

$$\therefore \tan \beta = \frac{h+x}{\frac{h}{\tan \alpha}}$$

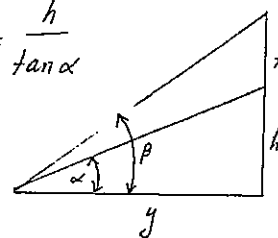
$$\tan \beta = \frac{h \tan \alpha + x \tan \alpha}{h}$$

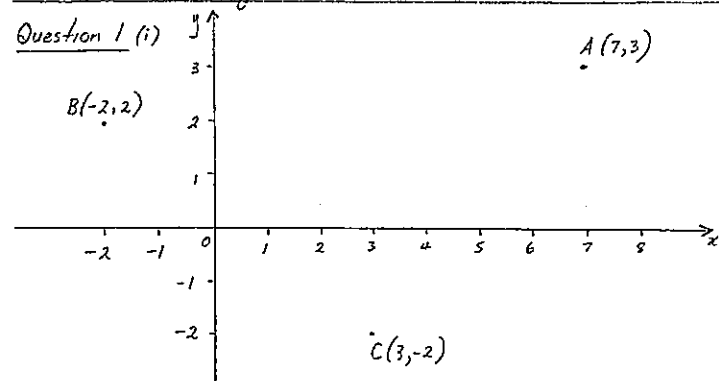
$$h \tan \beta = h \tan \alpha + x \tan \alpha$$

$$h \tan \beta - h \tan \alpha = x \tan \alpha$$

$$h(\tan \beta - \tan \alpha) = x \tan \alpha$$

$$h = \frac{x \tan \alpha}{\tan \beta - \tan \alpha}$$





$$\therefore x - 21 = 0$$

$$x = 21$$

D has coordinates (21,0)

$$vi) d = \left| \frac{1(7) + (-9)(3) - 21}{\sqrt{1^2 + (-9)^2}} \right| \text{ using } d = \left| \frac{ax_1 + by_1 + c}{\sqrt{a^2 + b^2}} \right|$$

$$d = \left| \frac{7 - 27 - 21}{\sqrt{82}} \right| \quad \text{and } a = 1 \quad x_1 = 7$$

$$b = -9 \quad y_1 = 3$$

$$c = -21$$

$$d = \left| \frac{-41}{\sqrt{82}} \right|$$

$$d = \frac{41}{\sqrt{82}}$$

$$vii) d_{AB} = \sqrt{(7+2)^2 + (3-2)^2}$$

$$= \sqrt{82}$$

$$\text{Area of } \triangle ABD = \frac{1}{2}bh$$

$$= \frac{1}{2} \times \sqrt{82} \times \frac{41}{\sqrt{82}}$$

$$= 20.5 \text{ sq units}$$

$$(ii) \text{ Midpoint of AC} = \left(\frac{7+3}{2}, \frac{3-2}{2} \right) \text{ using } \left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2} \right)$$

$$= \left(5, \frac{1}{2} \right)$$

$$iii) \text{ Gradient of AB} = \frac{3-2}{7+2} \text{ using } m = \frac{y_2-y_1}{x_2-x_1}$$

$$m_{AB} = \frac{1}{9}$$

$$iv) \text{ Equation of L is :-}$$

$$y+2 = \frac{1}{9}(x-3) \quad \text{using } y-y_1 = m(x-x_1)$$

$$9y+18 = x-3$$

$$x-9y-21=0$$

Question 2

$$a) 6x+2y-7=0$$

$$2y = -6x+7$$

$$y = -3x + \frac{7}{2}$$

$\therefore$ Gradient of perpendicular line is $\frac{1}{3}$

$$b) \text{ Gradient of line k} = \tan 150^\circ$$

$$= -\frac{1}{\sqrt{3}}$$

Equation of line k is

$$y-2 = -\frac{1}{\sqrt{3}}(x+\sqrt{3})$$

$$y\sqrt{3}-2\sqrt{3} = -x-\sqrt{3}$$

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$$c) x = 4-5y \rightarrow 2x-3y=5$$

$$2(4-5y)-3y=5$$

$$8-10y-3y=5$$

$$3 = 13y$$

$$y = \frac{3}{13}$$

$$x = 4 - \frac{15}{13}$$

$$x = 2\frac{11}{13}$$

Point of Intersection is $\left(2\frac{11}{13}, \frac{3}{13} \right)$

$$d) (3,2) \text{ satisfies } 4x+ay=b$$

$$12+2a=b \dots (1)$$

$$\text{Gradient of } 5x+2y=19 \text{ is } -\frac{5}{2} = m_1$$

$$\text{Gradient of } 4x+ay=b \text{ is } -\frac{4}{a} = m_2$$

$$\text{Since } \perp, m_1 \times m_2 = -1$$

$$\frac{20}{2a} = -1$$

$$20 = -2a$$

$$a = -10$$

$$\therefore b = 12-20 \text{ using (1)}$$

$$b = -8$$

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$$e) i) \text{ Equation of required line is}$$

$$(3x+4y-2)+k(5x-2y-12)=0$$

$$(1,-2) \text{ satisfies}$$

$$\therefore (3-8-2)+k(5+4-12)=0$$

$$-7-3k=0$$

$$k = -\frac{7}{3}$$